

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Shakthi S P
Program/Batch	B083-Networkwalks
Date	25 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes – activities were performed within the assigned training scope and on my own local network
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on systems and networks that were within the assigned cybersecurity training scope or on devices and networks that I own or am authorized to test. All these materials are for educational and research purposes only. These activities should not be used to gain unauthorized access to systems or networks. Any misuse of cybersecurity knowledge may have legal and professional consequences. The instructor, authors and Networkwalks are not responsible for unauthorized use of the information contained in this report.

2. Introduction

This report covers **footprinting and reconnaissance of the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and network discovery of my own local LAN using Zenmap (W2-PM5).** The footprinting activity focused on collecting publicly available information about the target domain using different reconnaissance tools. The network scanning activity focused on identifying active hosts, IP addresses, MAC addresses and network topology within my own local network. All commands for the footprinting activity were performed in **Kali Linux**, while **Zenmap and Windows CMD** were used for the local network scanning activity. Each activity was documented with the command used, the observed result, supporting evidence and a short explanation of its cybersecurity relevance.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Obtain publicly available domain registration and name-server information
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address
curl -I	Inspect HTTP response headers
wafw00f	Identify the Web Application Firewall protecting the website
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
Windows CMD	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools:

WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon.

Each tool was used to collect a different type of information about the target.

✓ WHOIS

I used **WHOIS** to obtain publicly available domain registration information and identify information associated with the domain and its name servers.

✓ WhatWeb

I used **WhatWeb** to identify the technologies used by the website. The results provided information about the web technologies and software exposed by the target.

✓ Nslookup

I used **nslookup** to resolve the domain name.

The result identified:

networkwalks.com → 192.232.216.135

This demonstrated that DNS resolution was working correctly and provided the IP address associated with the domain.

✓ Curl

I used the following command:

- `curl -I https://networkwalks.com`

The server returned:

- HTTP/2 200

The response also exposed HTTP header information and the WordPress REST API endpoint:

- `https://networkwalks.com/wp-json/`

This provided additional information about the web application.

✓ Wafw00f

I used **Wafw00f** to identify whether a Web Application Firewall was protecting the website.

The result identified:

ModSecurity (SpiderLabs) WAF

This indicates that a web application firewall is present in front of the website.

✓ DNSRecon

I used **DNSRecon** to enumerate DNS records for `networkwalks.com`.

The results provided information relating to:

- Name servers
- Mail server
- A record
- TXT/SPF information
- SRV records

- DNS infrastructure

The enumeration completed successfully and returned multiple DNS records.

4.2 Network Scanning with Zenmap

For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Windows `ipconfig` command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The results identified **five live hosts**:

- 192.168.1.1
- 192.168.1.9
- 192.168.1.12
- 192.168.1.20
- 192.168.1.21

The results also provided **five corresponding MAC addresses** for the live hosts. The MAC address information was recorded as part of the practical and is included in the supporting evidence.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified Apache, WordPress, WordPress Download Manager, jQuery, Bootstrap and other web technologies	Attackers may use exposed technology information to identify software requiring further security review	● Medium
2	Server IP address identifiable	Nslookup resolved networkwalks.com to 192.232.216.135	Provides information that can contribute to understanding the target infrastructure	● Low
3	HTTP technical information exposed	Curl returned HTTP/2 response headers, Apache server information, cookie information and a WordPress REST API link	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs) WAF	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified NS, A, MX, TXT and SRV records, with 8 records found	DNS information can help build a broader infrastructure profile	● Medium
6	Multiple live hosts visible on local network	Zenmap identified five live hosts in the local network	Unknown or unauthorized devices may potentially be present on a network	● Medium

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software technology, network address, HTTP header, DNS record or live host does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated**
CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
The identified ModSecurity WAF should be kept enabled, properly configured and regularly monitored.
6. **Perform regular internal network discovery**
Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

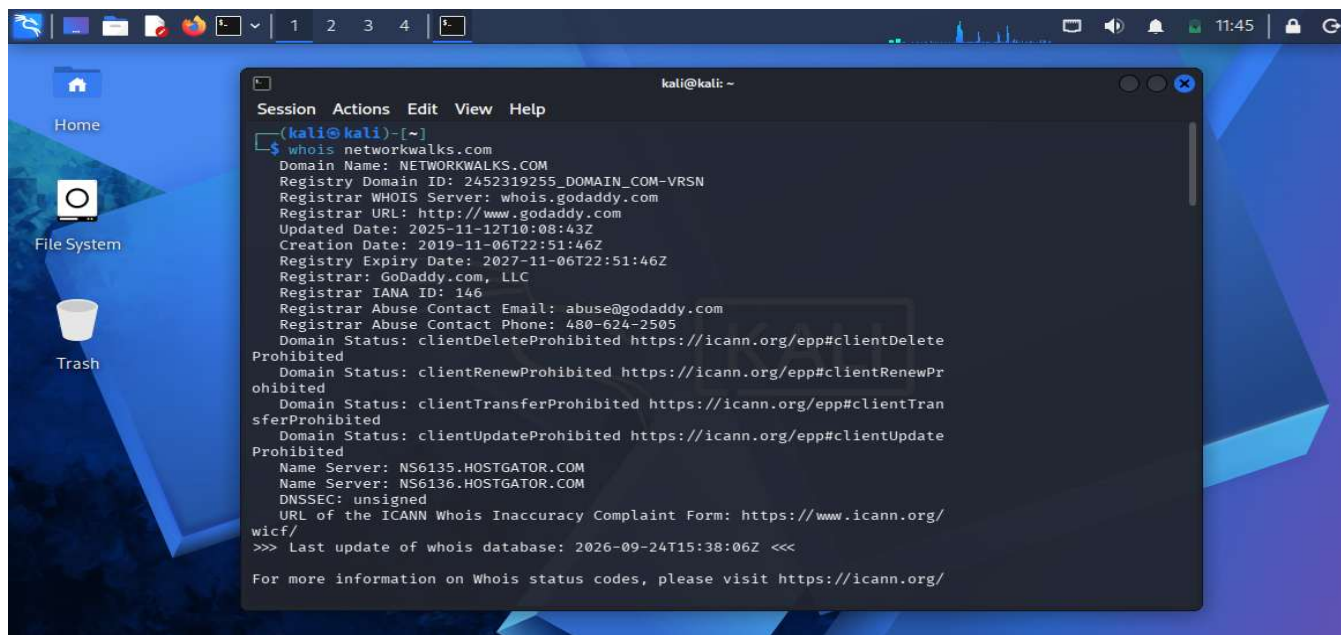
7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning. In the footprinting activity, I used six Kali Linux tools to collect information about the assigned target domain. I learned how **WHOIS** can provide domain registration information, **WhatWeb** can identify web technologies, **NSLookup** can perform DNS resolution, **Curl** can inspect HTTP response headers, **Wafw00f** can identify a Web Application Firewall, and **DNSRecon** can enumerate DNS records. In the network scanning activity, I used **Zenmap** to identify my local network configuration and discover active hosts. The Ping Scan identified **five live hosts**, and I also collected their IP and MAC address information as required by the practical. I then generated and saved the network topology. The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses. I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

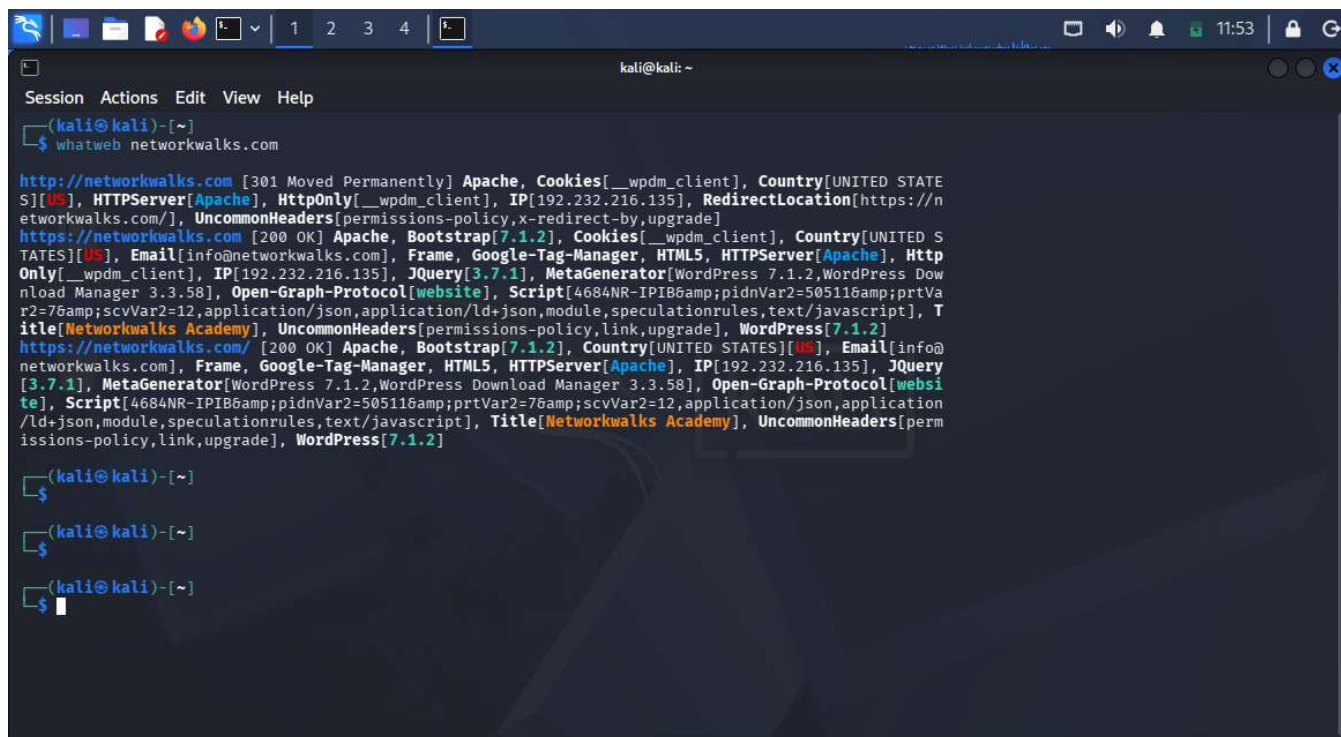
8. Evidences Collected

Figure 1 — WHOIS Output



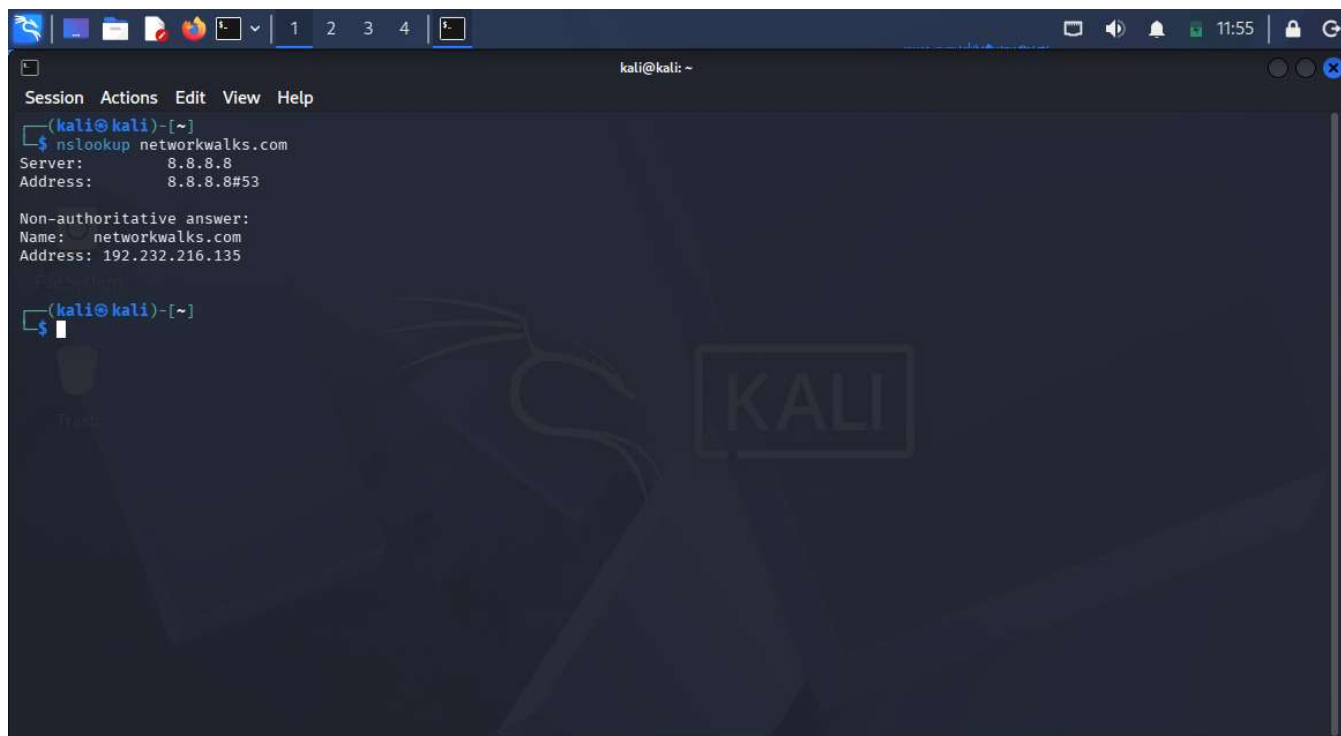
```
kali@kali: ~  
Session Actions Edit View Help  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-09-24T15:38:06Z <<<  
For more information on Whois status codes, please visit https://icann.org/
```

Figure 2 — WhatWeb Output



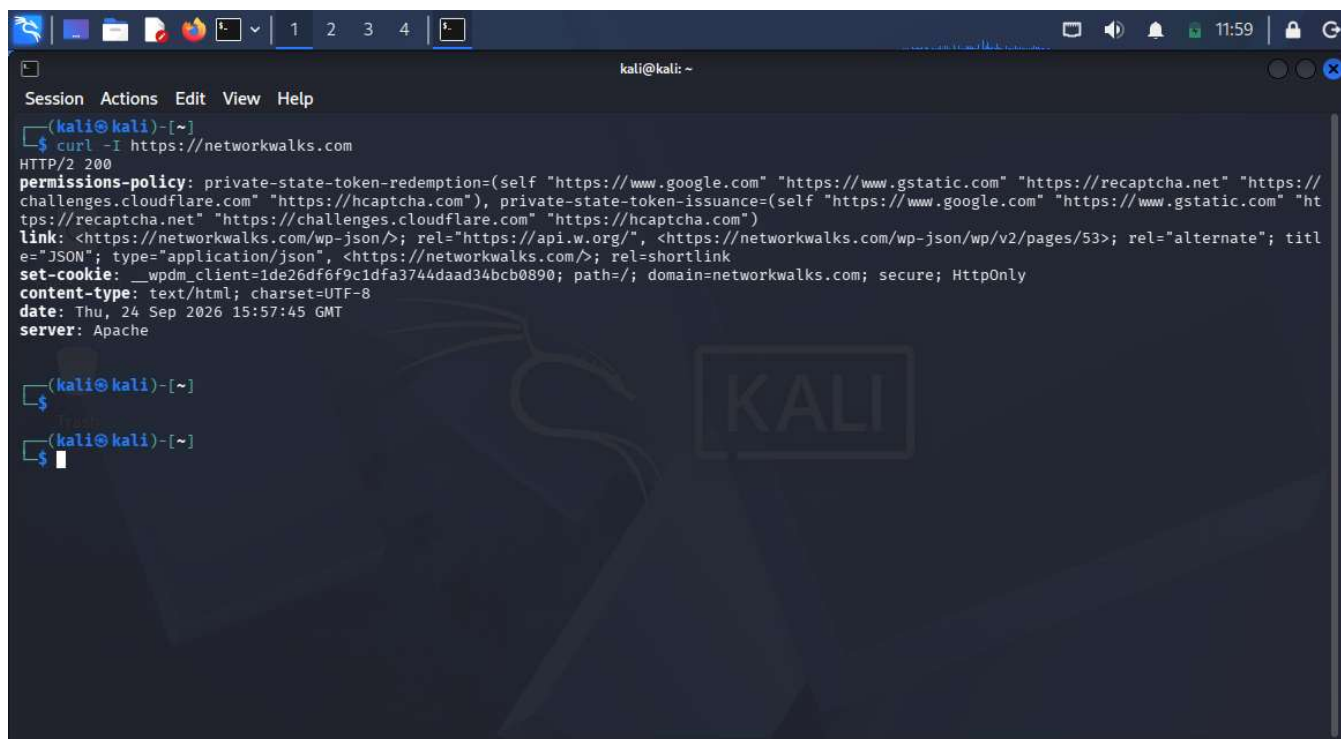
```
kali@kali: ~  
Session Actions Edit View Help  
$ whatweb networkwalks.com  
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[_wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade]  
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.2], Cookies[_wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.2,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB6amp;pidnVar2=505118amp;prtVar2=76amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade], WordPress[7.1.2]  
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.2], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.2,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB6amp;pidnVar2=505118amp;prtVar2=76amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade], WordPress[7.1.2]  
  
$  
$  
$
```

Figure 3 — NSLookup Output



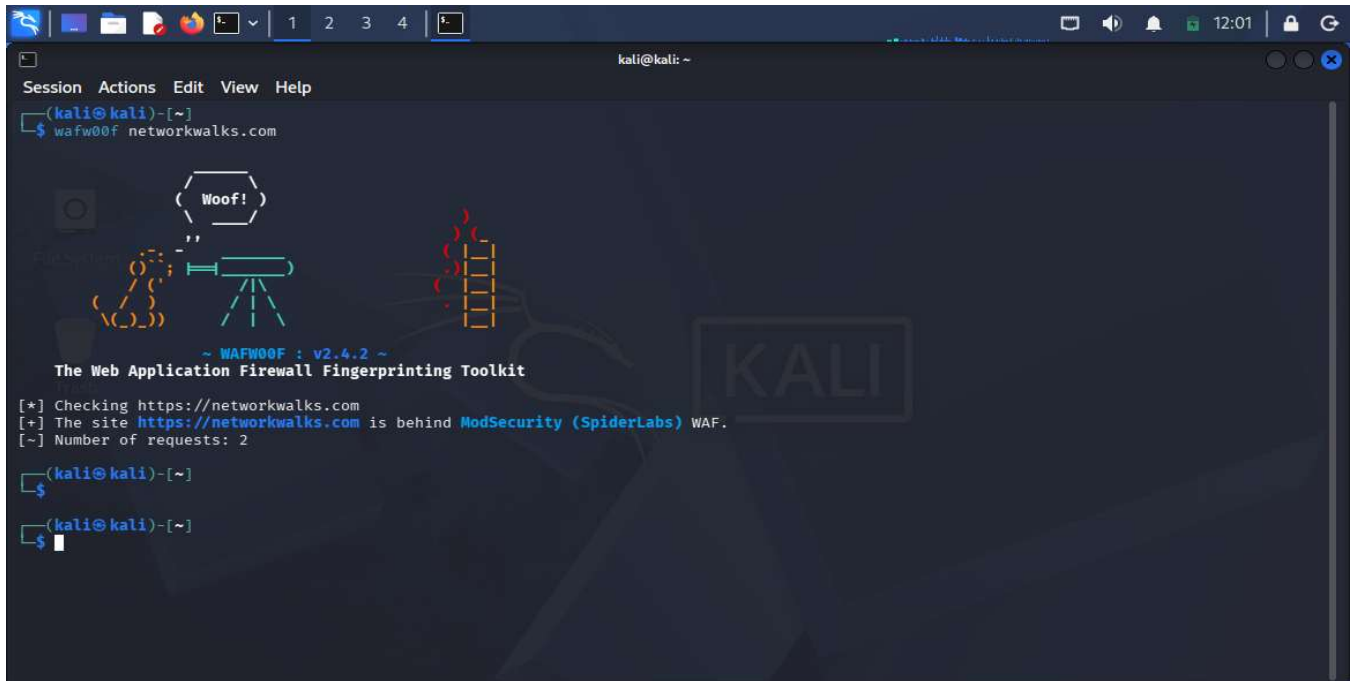
```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ nslookup networkwalks.com  
Server:      8.8.8.8  
Address:     8.8.8.8#53  
  
Non-authoritative answer:  
Name:   networkwalks.com  
Address: 192.232.216.135  
  
(kali@kali)-[~]  
$
```

Figure 4 — Curl Output



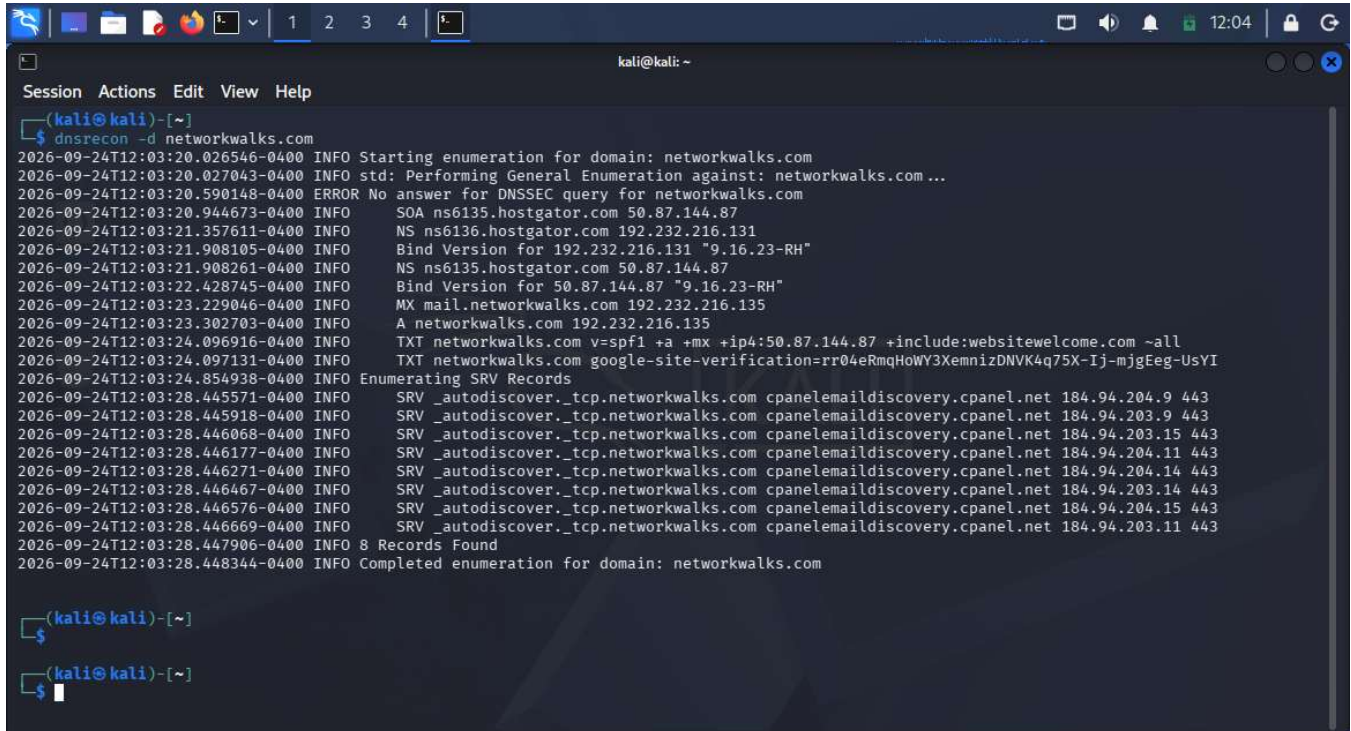
```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ curl -I https://networkwalks.com  
HTTP/2 200  
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")  
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel="shortlink"  
set-cookie: __wpdm_client=1de26df6f9c1dfa3744daad34bcb0890; path=/; domain=networkwalks.com; secure; HttpOnly  
content-type: text/html; charset=UTF-8  
date: Thu, 24 Sep 2026 15:57:45 GMT  
server: Apache  
  
(kali@kali)-[~]  
$
```

Figure 5 — Wafw00f Output



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ wafw00f networkwalks.com  
  
~ WAFW00F : v2.4.2 ~  
The Web Application Firewall Fingerprinting Toolkit  
[*] Checking https://networkwalks.com  
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.  
[~] Number of requests: 2  
  
(kali@kali)-[~]  
$  
  
(kali@kali)-[~]  
$
```

Figure 6 — DNSRecon Output



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ dnsrecon -d networkwalks.com  
2026-09-24T12:03:20.026546-0400 INFO Starting enumeration for domain: networkwalks.com  
2026-09-24T12:03:20.027043-0400 INFO std: Performing General Enumeration against: networkwalks.com...  
2026-09-24T12:03:20.590148-0400 ERROR No answer for DNSSEC query for networkwalks.com  
2026-09-24T12:03:20.944673-0400 INFO SOA ns6135.hostgator.com 50.87.144.87  
2026-09-24T12:03:21.357611-0400 INFO NS ns6136.hostgator.com 192.232.216.131  
2026-09-24T12:03:21.908105-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"  
2026-09-24T12:03:21.908261-0400 INFO NS ns6135.hostgator.com 50.87.144.87  
2026-09-24T12:03:22.428745-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"  
2026-09-24T12:03:23.229046-0400 INFO MX mail.networkwalks.com 192.232.216.135  
2026-09-24T12:03:23.302703-0400 INFO A networkwalks.com 192.232.216.135  
2026-09-24T12:03:24.096916-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com -all  
2026-09-24T12:03:24.097131-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI  
2026-09-24T12:03:24.854938-0400 INFO Enumerating SRV Records  
2026-09-24T12:03:28.445571-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443  
2026-09-24T12:03:28.445918-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443  
2026-09-24T12:03:28.446068-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443  
2026-09-24T12:03:28.446177-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443  
2026-09-24T12:03:28.446271-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443  
2026-09-24T12:03:28.446467-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443  
2026-09-24T12:03:28.446576-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443  
2026-09-24T12:03:28.446669-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443  
2026-09-24T12:03:28.447906-0400 INFO 8 Records Found  
2026-09-24T12:03:28.448344-0400 INFO Completed enumeration for domain: networkwalks.com  
  
(kali@kali)-[~]  
$  
  
(kali@kali)-[~]  
$
```

Figure 7 — Zenmap Scan Results

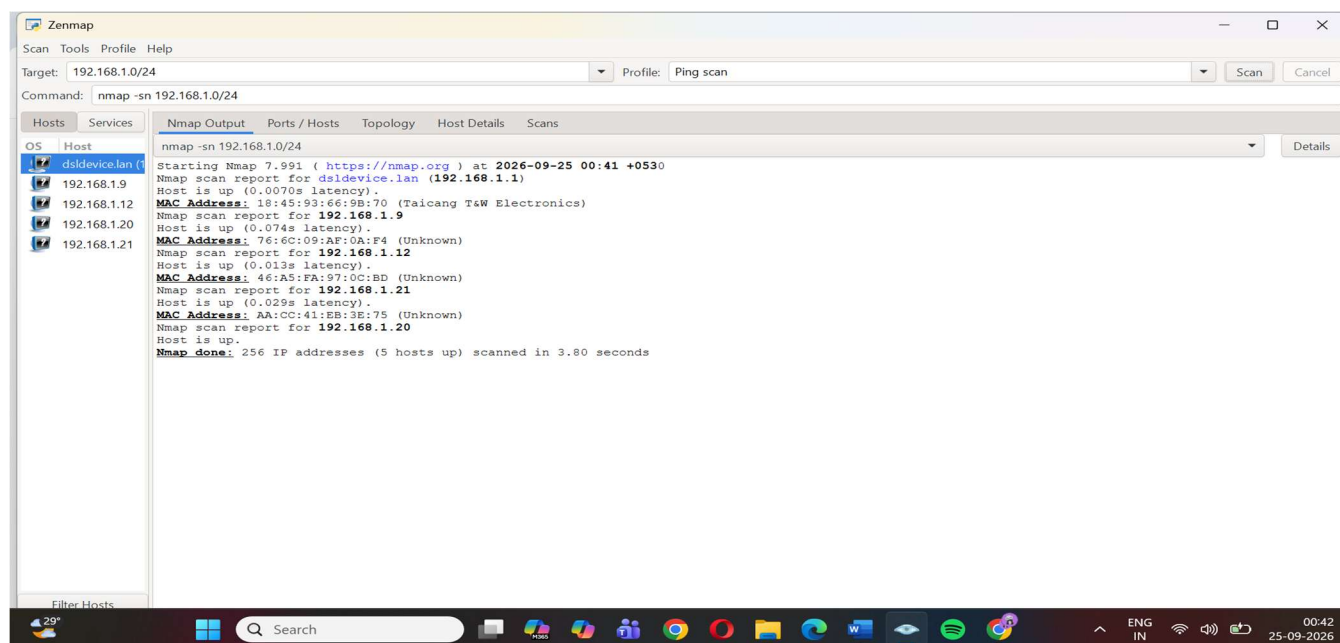
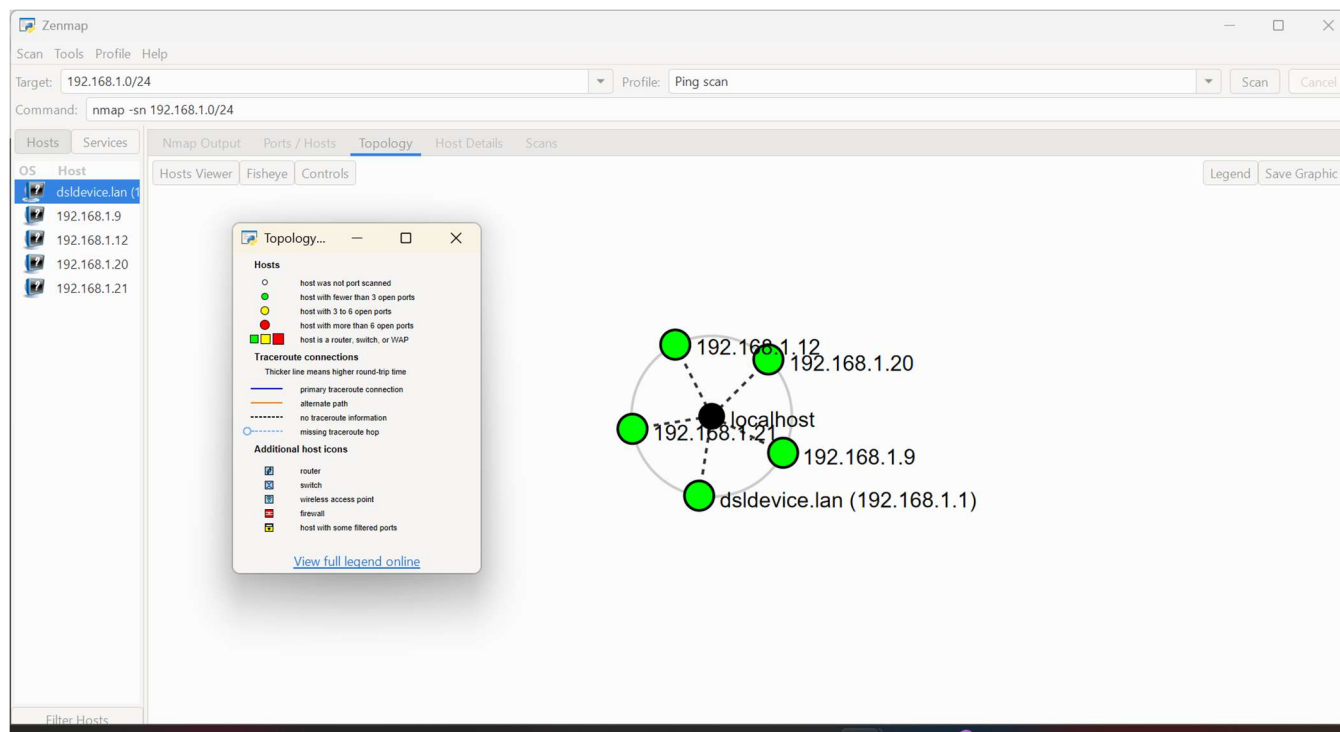


Figure 8 — Zenmap Network Topology



-End-



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Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub